

Visual Dictionary

Sprint: Programming the Web

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PROJECT SCOPE

Team

- 3-4 members collaborating on design, coding, and analysis
- Shared roles:
 - Sohail : formatting and css
 - Anurag : api call and fixed bugs
 - Anil : designing

Goals

- ✓ Build an interactive webpage using 2 public APIs
- ✓ Implement DOM manipulation and event listeners
- ✓ Analyze browser-server communication using Wireshark

Tools & Technologies

- HTML, CSS, JavaScript (Fetch API, Async/Await)
- Wireshark for network packets tracing
- GitHub Pages / Netlify / Vercel for deployment

APIs Used

Wikipedia API - Fetches summaries and thumbnail images for searched words.

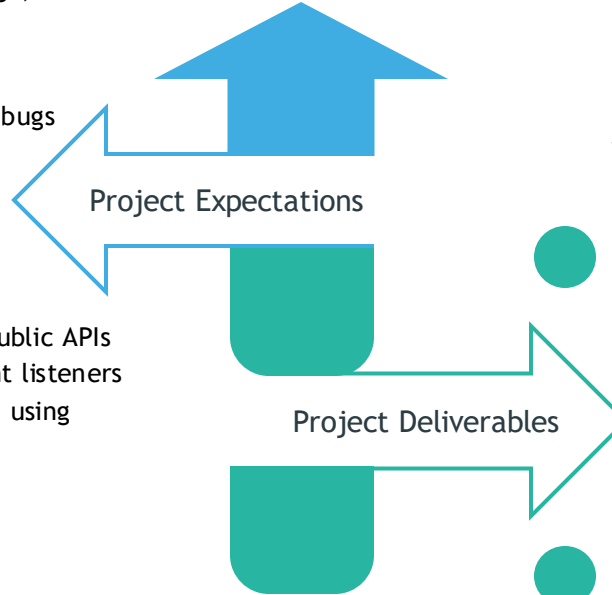
Web Speech API - Converts displayed text to speech.

Deliverables

- ✓ Hosted webpage link
- ✓ Source code (index.html, style.css, script.js)
- ✓ Wireshark captures + 2-3 page analysis report

Constraints

- Use only open/free APIs
- Page must show clear DOM updates & interactivity
- Hosted version must load without any dependencies



PROJECT SCOPE & DELIVERABLES

Phase

Phase 1: Design & Setup

Phase 2: Development & Integration

Phase 3: Testing, Analysis & Hosting

Scope

Wikipedia API: fetch article content and information from Wikipedia programmatically.
Web Speech API: Enables speech-to-text and text-to-speech functionality directly.

A quick-info search site with images, powered by the Wikipedia API.
Enter a word in the search bar and get quick info with an image below.

We fetch data from the Wikipedia API using JavaScript's `fetch()` method. The returned image and summary are then dynamically shown by updating the corresponding DOM elements, triggered through a search button event listener.

Deliverables

APIs: Wikipedia API & Web Speech API.

Design: Simple UI with searchbar image and text description with options for toggle theme

- Working prototype showing API data updates
- Code files (`index.html`, `style.css`, `script.js`)
- Screenshots of web page functionality

- Wireshark capture (.pcapng or screenshots)
- Visual Dictionary
- 2-3 page report with screenshots and findings